Roman IV Canlas

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SUMMARY

Computer Science student specializing in AI Engineering and Data Science, with hands-on experience building production-grade RAG architectures, real-time generative AI streaming backends, and predictive spatial models. Proven ability to engineer end-to-end systems—from custom DSL parsers and FastAPI WebSockets to deterministic AI safety guardrails and normalized PostgreSQL schemas.

EDUCATION

Bachelor of Science in Computer Science |Angeles University Foundation| 2023 - Present

Relevant Coursework: Machine Learning, Artificial Intelligence, Compiler Design / Programming Languages, Database Management Systems, Data Structures & Algorithms

TECHNICAL SKILLS

- **AI & Machine Learning**: Python, LangChain, LangGraph, FastMCP, Google Gemini & Lyria APIs (google-genai), Scikit-learn, TF-IDF, Computer Vision (COCO-SSD), MaxEnt
- **Backend, Data & Databases**: FastAPI, WebSockets, Flask, Node.js, PostgreSQL, MySQL, SQL, Pandas, NumPy, MSPA
- **Mobile, Frontend & Tools**: Flutter, Dart, JavaScript (ES6), HTML/CSS, Git, GitHub Actions, Google Colab, Figma

AI ENGINEERING & DATA SCIENCE PROJECTS

StockPal: Smart Inventory Management System | Python, Flask, PostgreSQL, SQL | Feb 2026 – May 2026

- Co-developed a full-stack inventory platform over a 60-day sprint, designing a normalized PostgreSQL database schema, Flask REST API endpoints, and automated CSV reporting pipelines.

SonataPy: Real-Time AI Music Generator & DSL Interpreter | Python, FastAPI, WebSockets, Google Lyria (google-genai), JS| Nov 2025

- Built a real-time generative audio streaming server using Python FastAPI, Uvicorn, and WebSockets (ws://) to stream continuous AI-generated music via Google's Lyria model.

Rodent Habitat Suitability & Connectivity Modeling (Thesis) | Python, MaxEnt, MSPA | Feb 2026 – Oct 2026

- Engineered species distribution models (SDMs) across Luzon, Philippines using MaxEnt to evaluate habitat suitability across [16] environmental and bioclimatic variables.

LANGUAGE SKILLS

- **Filipino**
- **English**

CERTIFICATIONS

- **Cisco Introduction to Data Science**
- **Cisco Introduction to Cybersecurity**